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Chapter 12, Problem 4SPA | 4 Bookmarks | Show all steps: ON

Step-by-step solution

Step 1 of 10

A perfectly competitive firm's supply curve is made up of the marginal cost curve at all prices above minimum average variable cost and the vertical axis at all prices below minimum average variable cost.

Comment

Step 2 of 10

Following table shows the *TC* , *AVC* and *MC* schedule of Pat's Pizzas Kitchen:

Output (pizzas per hour)	Total cost (dollars per hour)	Average variable cost (dollars per pizza)	Marginal cost (dollars per pizza)
0	10	0	11
1	21	11	9
2	30	10	11
3	41	10.33	13
4	54	11	15
5	69	11.8	

Comment

Step 3 of 10

Marginal cost of increasing output from 0 to 1 pizza per hour is $11 and $9 so the marginal cost of the First pizza is half – way between $11 and $9, which is $10 per pizza. Thus, the marginal cost of producing 1 pizza is $10.

Comment

Step 4 of 10

Marginal cost of increasing output from 2 to 3 pizzas per hour is $11 and $13 so the marginal cost of the third pizza is half – way between $11 and $13, which is $12 per pizza. Thus, the marginal cost of producing 3 pizzas is $12.

Comment

Step 5 of 10

Marginal cost of increasing output from 3 to 5 pizzas per hour is $13 and $15 so the marginal cost of the fourth pizza is half – way between $13 and $15, which is $14 per pizza. Thus, the marginal cost of producing 4 pizzas is $14.

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In perfect competition, firm's supply curve is derived from *MC* curve at prices above minimum average variable cost. *AVC* is at its minimum when it equals *MC*.

Comment

Step 7 of 10

The marginal cost of producing 2 pizzas is $10, while average variable cost of producing 2 pizzas is also $10. So, when Pat's Pizza Kitchen produces 2 pizzas, its *MC* is equal to its *AVC*. So, the Pat's supply schedule would be same as that of *MC* schedule for 2 pizzas per hour or more.

Comment

Step 8 of 10

Following table shows the *MC* schedule of Pat's Pizza Kitchen when it produces 2 Pizzas or more:

Output (pizzas per hour)	Marginal cost (dollars per pizza)
2	9
3	11
4	13

Comment

Step 9 of 10

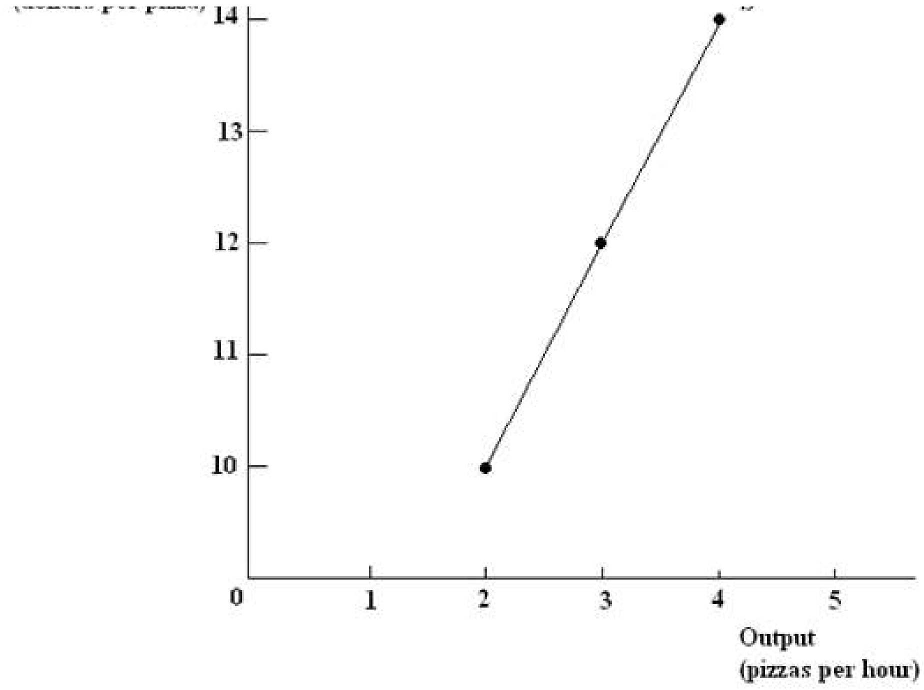
Following is the supply schedule of Pat's Pizza Kitchen derived from above *MC* schedule:

Price (dollars per pizza)	Supply (output per hour)
10	2
12	3
14	4

Comment

Step 10 of 10

Following figure shows the supply curve of Pat's Pizza Kitchen:



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Step-by-step solution

Step 1 of 7

Suppose there are 1000 perfectly competitive firms in the paper market. The market demand schedule and each firm's cost schedule are given.

Comment

Step 2 of 7

Market price is the found when the supplied quantity is equal to the demanded quantity in the market.

By merging both of the table, the new table so formed is as follows:

Price (dollars per box)	Quantity demanded (thousands of boxes per week)	Output (boxes per week)	Marginal Cost (dollars per additional box)	Average Variable Cost (dollars per box)	Average Total Cost (dollars per box)
3.65	500	200	6.40	7.80	12.80
5.20	450	250	7.00	7.00	11.00
6.80	400	300	7.65	7.10	10.43
8.40	350	350	8.40	7.20	10.06
10.00	300	400	10.00	7.50	10.00
11.60	250	450	12.40	8.00	10.22
13.20	200	500	20.70	9.00	11.00

Comment

Step 3 of 7

Market price is obtained when the quantity demanded is equal to the quantity supplied. In the above table quantity demanded is given in second column. Since firms are perfectly competitive, the supply will be same as marginal cost above minimum average variable cost.

In the above table minimum average variable cost is 7.00. Thus, the supply of each firm is equal to quantity of 250 boxes and above.

Therefore, after observing the above table, the market price will be equal to \$8.40, the reason being the definition of the market price. As it would be clearly observed that the level of quantity demanded (i.e. 350 boxes) at the above stipulated price is equal to the supply (i.e. 350 boxes) and thus the equilibrium price or market price is obtained.

Hence, the market price is $\boxed{\$8.40}$.

Comment

Step 4 of 7

At the price level of \$8.40, the quantity supplied by each firm is 350 boxes. Since there are 1000 firms in the market, the market output will be equal to the quantity supplied at equilibrium price multiply by 1,000 firms.

Hence, the market output is $\boxed{350{,}000 \text{ boxes}}$.

Comment

Step 5 of 7

The quantity produced by each firm is total market output divided by the number of firms, that is, 350 boxes $\left[=\frac{350{,}000}{1000}\right]$.

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Comment

Step 7 of 7

Each firm sell 350 boxes at a market price of $8.40 and produce 350 boxes at the average total cost of $10.06. Since the average total cost is greater than the market price, each firm will earn economic loss.

The economic loss or profit for each firm is computed below:

$$\begin{aligned}\text{Economic loss} &= (P - AC) \times Q \\ &= (\$8.40 - \$10.06) \times 350 \\ &= -\$1.66 \times 350 \\ &= -\$581\end{aligned}$$

Hence, the economic loss for each firm is $\boxed{-\$581}$.

Comment

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Step-by-step solution

Step 1 of 2

"Market price is the found when the supplied quantity is equal to the demanded quantity in the market."

Since the costs to the output will remain same so the comparison has to be done with new price level and the demand for the quantity.

It is also known that the market price is determined when quantity demanded is equal to the quantity supplied.

By merging both of the table, the new table so formed is as follows:

	Average Variable Cost	Average Total Cost			
Price (dollars per box)	Quantity demanded (thousands of boxes per week)	Output (thousands of boxes per week)	Marginal Cost (dollars per additional box)	(dollars per box)	
2.95	500	200	6.40	7.80	12.80
4.13	450	250	7.00	7.00	11.00
5.30	400	300	7.65	7.10	10.43
6.48	350	350	8.40	7.20	10.06
7.65	300	400	10.00	7.50	10.00
8.83	250	450	12.40	8.00	10.22
10.00	200	500	20.70	9.00	11.00
11.18	150	-	-	-	-

a.

Observing the above table, the demand and supply is equal at the quantity of 350 boxes. Therefore the market price is $6.48 in this case.

Answer is $\boxed{\$6.48.}$

Comment

Step 2 of 2

b.

In order to find the profit and loss of the firm, the price and average total cost at a given quantity has to be known. The formula representing the profit or loss is as follows:

$$\text{Profit or loss} = (P - ATC) \times Q$$

The price for the mentioned level of output is $6.48, average total cost for the level of output is $10.06 and the level of output is 350 boxes. Therefore the profit or loss for the firm is as follows:

$$\text{Loss} = (6.48 - 10.06) \times 350$$
$$\text{Loss} = -3.58 \times 10.06$$
$$\text{Loss} = -\$1{,}253$$

Therefore the firm will experience the economic loss which will be equal to $\boxed{-\$1{,}253.}$

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Step-by-step solution

Step 1 of 5

The perfect competition is the market where there are large numbers of buyers and sellers, selling the identical products with price as constant. Both the buyers and sellers have perfect knowledge about the market.

Comment

Step 2 of 5

There is a perfectly competitive market for papers with the given demand schedule and cost schedule. It is needed to calculate the market price and quantity produced in the long run.

In the long run, the equilibrium of the perfectly competitive firm will exist at a point where price must equal average total cost. It can be seen at $10 per box. When the firm decides to exit the market, the prices increases. At the above equilibrium condition, the output is 400 boxes per week.

Comment

Step 3 of 5

The total quantity produces at $10 per box can be calculated as follows:

$$\begin{aligned} Q_D &= 400 \times \text{total number of firms} \\ &= 400 \times 1{,}000 \\ &= 400{,}000 \end{aligned}$$

Hence, the total demand of paper is $\boxed{400{,}000}$.

Comment

Step 4 of 5

The number of firms (N) can be calculated using the following formula:

$$N = \frac{\text{Total}\, Q_D}{\text{output produced by each firm}}$$

The total quantity demanded at $10 per box can be calculated as follows:

$$\begin{aligned} Q_D &= 300 \times \text{total number of firms} \\ &= 300 \times 1{,}000 \\ &= 300{,}000 \end{aligned}$$

Thus, the total demand of paper is 300,000.

Comment

Step 5 of 5

Now, substituting the calculated values in the formula for determining the number of firms, the number of firms can be determined as follows:

$$\begin{aligned} \text{Number of firms} &= \frac{3{,}00{,}000}{400} \\ &= 750 \end{aligned}$$

Hence, the price of paper produced is $\boxed{\$10}$, quantity produced is $\boxed{400{,}000\,\text{boxes}}$ and the number of firms in the market is $\boxed{750}$.

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Step-by-step solution

Step 1 of 2

The perfect competition is the market where there are large numbers of buyers and sellers selling the identical products with price as constant. Both the buyers and sellers have perfect knowledge about the market.

Comment

Submit

Step 2 of 2

Using the new demand schedule and cost schedule given in the textbook, it is needed to calculate the market price, output and economic loss or profit.

The equilibrium condition for the perfectly competitive market occurs at the point where price equals average total cost. It can be seen at $10 per box. When the firm decides to exit the market, the prices increases. At the above equilibrium condition, the output is 200 boxes per week.

The total quantity demanded at $10 per box can be calculated as done below:

$$\begin{aligned} Q_D &= 200 \times \text{total number of firms} \\ &= 200 \times 1{,}000 \\ &= 200{,}000 \end{aligned}$$

Thus, the market output is 200,000 boxes.

The economic profit and loss will be zero for each firm because in the long run, price always equals average total cost which implies no profit no loss situation.

Hence, the price of paper produced is $\boxed{\$10}$,quantity produced is $\boxed{200{,}000\,\text{boxes}}$.

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Step-by-step solution

Step 1 of 5

Perfect Competition is a market in which following conditions prevail:

- Many firms sell identical products to many buyers.
- There are no restrictions on entry into the market.
- Established firms have no advantage over the new ones.
- Sellers and buyers are well informed about prices.

Comment

Step 2 of 5

Following conditions prevail in the market in which these gas stations operate:

- Many other gas stations sell gasoline that is identical to the gasoline sold by these gas stations.
- There is no restriction on entry into the market for gasoline.
- Buyers and sellers are well informed about prices.

Comment

Step 3 of 5

Since, conditions prevailing in the market in which these gas stations operate are similar to the conditions that prevail in perfectly competitive market.

Therefore, these gas stations operate in a perfectly competitive market.

Comment

Step 4 of 5

In a **Perfectly Competitive Market**, market demand and market supply determines the price of a good. Since, market for gasoline is also a perfectly competitive market, market demand for gasoline and market supply of gasoline will determine the price of gasoline. Price at which market demand for gasoline is equal to the market supply of gasoline would be the price of gasoline.

Comment

Step 5 of 5

In perfect competition, firm's marginal revenue equals the market price. Since, these gas stations also operate in perfectly competitive market, so marginal revenue from gasoline will be determined by market price of gasoline.

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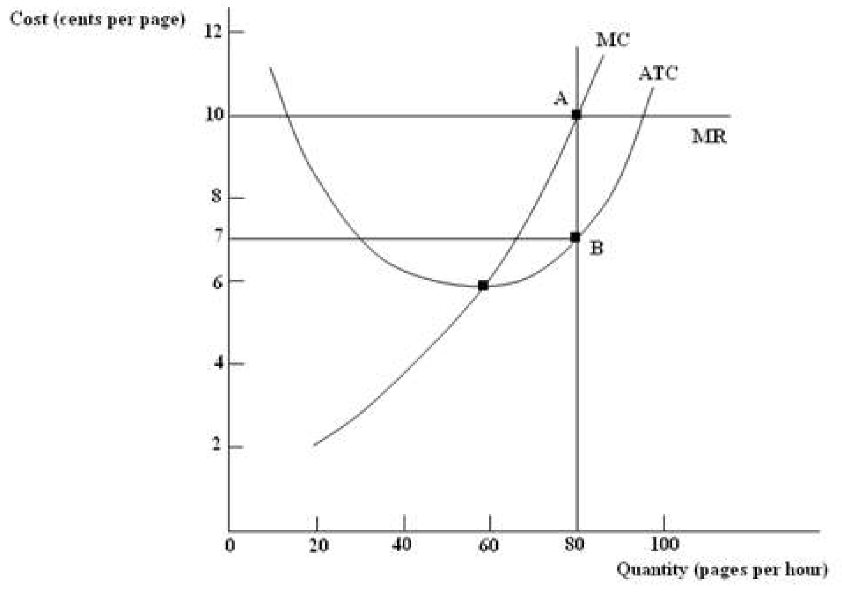
Step-by-step solution

Step 1 of 4

Following figure shows the costs of Quick Copy, one of many copy shops near campus:



Comment

Step 2 of 4

In **Perfect Competition**, the firm's marginal revenue equals the market price. Since Quick Copy is one of the many shops near the campus, so quick copy is operating in a perfectly competitive market and would be a perfectly competitive firm. So, Quick Copy's marginal revenue would also be equal to the market price of copying. It is given that market price of copying is 10 cents a page. So, marginal revenue of Quick Copy would also be 10 cents per page and as above figure shows that marginal revenue curve of Quick Copy would be a straight horizontal line at the market price of 10 cents per page.

Comment

Step 3 of 4

(a) Quick Copy's profit-maximizing output is 80 pages per hour.

Explanation: A firm in perfectly competitive market maximizes its profit by producing the output at which marginal revenue equals marginal cost and marginal cost is increasing. So, Quick Copy being a perfectly competitive firm would also maximize its profit by producing the output at which its marginal revenue is equal to its marginal cost and its marginal cost is increasing. As the figure shows that at point A, Quick Copy's MR is equal to its MC and also MC is increasing. So, output corresponding to point A would be the profit maximizing output, which is 80 pages per hour.

Comment

Step 4 of 4

(b) Production of Quick Copy is 80 pages per hour. For producing 80 pages per hour, Quick Copy incurs *ATC* of 7 cents per page.

Average total cost is total cost per unit of output.

$\text{ATC} = \text{TC} \div \text{Q}$.

Rearranging the above equation,

$\text{TC} = \text{ATC} \times \text{Q}$

Then,

$\text{TC} = 7 \text{ cents} \times 80 \text{ pages}$

$\text{TC} = 560 \text{ cents or } \$5.60$

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$= 800 \text{ cents or } \$8.00$

Hence,

$$\text{Economic Profit} = TR - TC$$
$$= \$\$8.00 - \$5.60$$
$$= \boxed{\$2.40}$$

So, Quick Copy will be making an economic profit of $2.40.

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Step-by-step solution

Step 1 of 7

Following table shows the market demand schedule for smoothies:

Table (a):

Price	Quantity demanded
(dollars per smoothie)	(smoothies per hour)
1.90	1,000
2.00	950
2.20	800
2.91	700
4.25	550
5.25	400
5.50	300

Comment

Step 2 of 7

Following table shows the cost schedule of each producer of smoothie:

Table (b):

Output	Marginal cost	Average variable cost	Average total cost
(smoothies per hour)	(dollars per additional smoothie)	(dollars per smoothie)	(dollars per smoothie)
3	2.50	4.00	7.33
4	2.20	3.53	6.03
5	1.90	3.24	5.24
6	2.00	3.00	4.67
7	2.91	2.91	4.34
8	4.25	3.00	4.25
9	8.00	3.33	4.44

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Step 3 of 7

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Step 4 of 7

In perfectly competition, firm's supply curve is derived from *MC* curve at prices above minimum average variable cost. *AVC* is at its minimum when it equals *MC*. As above table (b) shows when firm produces 7 smoothies per hour its *MC* is equal to its *AVC*.

So, the firm's supply schedule would be same as that of *MC* schedule for 7 smoothies per hour or more. In perfect competition, *MC* is equal to price. So, each value of *MC* also represents the price at which firm is willing to supply the corresponding output. For example, at the *MC* or price of $2.91 per smoothie, a firm supplies 7 smoothies per hour. There are 100 firms in the market. So, total supply in the market would be 700 smoothies per hour. The quantity demanded at $2.91 per smoothie is also 700 smoothies per hour {table (a)}.

Since, at $2.91 per smoothie, market demand is equal to the market supply, therefore $2.91 per smoothie is the market price.

Comment

Step 5 of 7

b. Market output is equal to the quantity supplied at market price. At market price of $2.91 per smoothie, 700 smoothies per hour are supplied in the market.

So market output is 700 smoothies per hour.

Comment

Step 6 of 7

c. The market price is $2.91 per smoothie. As above table (b) shows that at market price of $2.91 per smoothie, each firm produces 7 smoothies per hour. So, output of each firm is 7 smoothies per hour.

Comment

Step 7 of 7

d. Production of each firm is 7 smoothies per hour. For producing 7 smoothies per hour, firm incurs ATC of $4.34 per smoothie.

Average total cost is total cost per unit of output.

$$\text{ATC} = \text{TC} \div \text{Q}$$

Rearranging the above equation,

$$\text{TC} = \text{ATC} \times \text{Q}$$

Then,

$$\text{TC} = 7 \times \$4.34$$
$$\text{TC} = \$30.38$$

$$\begin{aligned}\text{TR} &= \text{Market Price} \times \text{Output Supplied} \\ &= \$2.91 \times 7 \\ &= \$20.37\end{aligned}$$

Hence,

$$\begin{aligned}\text{Economic Profit} &= \text{TR} - \text{TC} \\ &= \$20.37 - \$30.38 \\ &= -\$10.01\end{aligned}$$

So, firm is incurring an economic loss of $10.01.

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Chap

Chapter 12, Problem 14APA

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Step-by-step solution

Step 1 of 3

a.

"Total cost is equal to total variable cost and total fixed cost in the short-run."

As the company takes the decision of shutdown, total fixed cost will not be changed. The reason is that company has to bear the cost of space, it is using for production. The cost of building also comes under fixed cost. In this regards even if there is no production, the company has to pay some fixed amount.

Therefore, the Total Fixed Cost will be $\boxed{\text{unchanged}}$ with the decision of shutdown.

Comment

Step 2 of 3

b.

Company takes the decision of shutdown when total revenue is less than the total cost of production. In this situation company bears the loss. In order to loss, the company bears the extra burden of total variable cost with increased unit of production. Since, total fixed cost has to be paid in any case therefore company start to cut their production.

Therefore, by cutting the supply of the product, the variable cost comes down with the unit and in this regard, total cost comes down. With decreased level of output company bears losses and revenue too.

Therefore by $\boxed{\text{decreasing the level of output}}$, 'Che Volt' minimizes its loss.

Comment

Step 3 of 3

c.

Production increases with the increase in demand for the product. Increased demand will escalate the prices of the product (i.e. automobile). Increased demand will encourage producers to increase the production for electric car. Increased prices will reduce the cost of production in a way that the company would experience the economies of scale.

In another way increased price will generate greater revenue for the company and thus in order to maximize the profit, the company will start production.

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Step-by-step solution

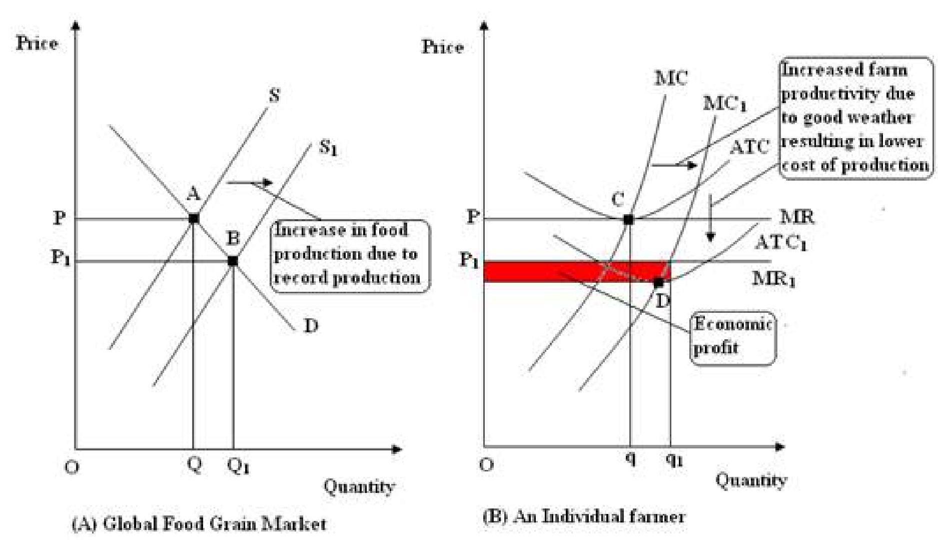
Step 1 of 6

In year 2008, food production has been at record high. The yield of food grains was better than expected. This better than expected yield or we can say record yield will increase the supply of food grains in the market. With demand remaining same, this increase in supply of food grains will bring a fall in price of food grains. Therefore, grain prices have fallen in 2008.

Comment

Step 2 of 6

Following figure shows the short-run effect of this fall in price of food grains on an individual farmer's economic profit:



Comment

Step 3 of 6

The food grain market starts out at long-run equilibrium. Panel (A) shows the long-run equilibrium of food grain market. Market is in equilibrium at point A where market demand curve D and market supply curve S intersects each other. Equilibrium market price is *OP* and the equilibrium market quantity is *OQ*. Panel (B) shows the situation of a representative farmer. Farmer is facing a marginal revenue curve *MR*, average total cost curve *ATC*, and marginal cost curve *MC*. Farmer is maximizing its profit by producing oq quantity of food grain. At market price, P, farmer is earning zero economic profit because the market price is equal to the minimum average total cost of the farmer (point C).

Comment

Step 4 of 6

Now, due to good weather conditions, record food grain production has taken place. With higher food grain yield, supply of food grain has increased. Due to increased supply of food grains market supply curve has shifted rightwards from S to S_1 (panel A). Market price has fallen from P to P_1 and Equilibrium market quantity has increased from Q to Q_1 (panel A).

Comment

Step 5 of 6

Decrease in market price has decreased the marginal revenue of the farmer as well and marginal revenue curve has shifted downward from *MR* to MR_1.

But good weather condition not only had increased the production of food grains but also has increased the farm productivity as well. This increased productivity resulted in decrease in cost of

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Step 6 of 6

Even if farmer is earning the economic profit in short run, increase in demand due to lower prices and entry of new farmers in the production of food grains due to incentive of economic profit will eventually return the farmer to zero economic profit scenario in the long run.

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Step-by-step solution

Step 1 of 8

Following table shows the market demand schedule for smoothies:

Table (a):

Price (dollars per smoothie)	Quantity demanded (smoothies per hour)
1.90	1,000
2.00	950
2.20	800
2.91	700
4.25	550
5.25	400
5.50	300

Comment

Step 2 of 8

Following table shows the cost schedule of each producer of smoothies:

Table (b):

Output (smoothies per hour)	Marginal cost (dollars per additional smoothie)	Average variable cost (dollars per smoothie)	Average total cost (dollars per smoothie)
3	2.50	4.00	7.33
4	2.20	3.53	6.03
5	1.90	3.24	5.24
6	2.00	3.00	4.67
7	2.91	2.91	4.34
8	4.25	3.00	4.25
9	8.00	3.33	4.44

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Step 3 of 8

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Step **4** of 8

In **Perfectly Competition**, firm's supply curve is derived from *MC* curve at prices above minimum average variable cost. *AVC* is at its minimum when it equals *MC*. As above table (b) shows when firm produces 7 smoothies per hour its *MC* is equal to its *AVC*.

So, the firm's supply schedule would be same as that of *MC* schedule for 7 smoothies per hour or more. In perfect competition, *MC* is equal to price. So, each value of *MC* also represents the price at which firm is willing to supply the corresponding output. For example, at the *MC* or price of $2.91 per smoothie, a firm supplies 7 smoothies per hour. There are 100 firms in the market. So, total supply in the market would be 700 smoothies per hour. The quantity demanded at $2.91 per smoothie is also 700 smoothies per hour {table (a)}.

Since, at $2.91 per smoothie, market demand is equal to the market supply, therefore $2.91 per smoothie is the market price.

Production of each firm is 7 smoothies per hour. For producing 7 smoothies per hour, firm incurs *ATC* of $4.34 per smoothie.

Average total cost is total cost per unit of output.

$$\text{ATC} = \text{TC} \div \text{Q}$$

Rearranging the above equation,

$$\text{TC} = \text{ATC} \times \text{Q}$$

Then,

$$\begin{aligned} \text{TC} &= 7 \times \$4.34 \\ \text{TC} &= \$30.38 \end{aligned}$$

$$\begin{aligned} \text{TR} &= \text{Market Price} \times \text{Output Supplied} \\ &= \$2.91 \times 7 \\ &= \$20.37 \end{aligned}$$

Hence,

$$\begin{aligned} \text{Economic Profit} &= \text{TR} - \text{TC} \\ &= \$20.37 - \$30.38 \\ &= \boxed{-\$10.01} \end{aligned}$$

So, firm is incurring an economic loss of $10.01.

Comment

Step **5** of 8

The long-run equilibrium market price of the firm in perfect competition is attained at a level where price equals minimum average total cost. At this equilibrium market price, firm makes only normal profit. If price falls below the minimum average total cost, then firm will incur economic loss in the long-run and will have to exit the market in order to eliminate the persistent economic losses.

Comment

Step **6** of 8

As the long-run equilibrium price of smoothies is less than the average total cost of the producers of smoothie, smoothie producers will be incurring an economic loss in the long-run. Since, existing firms in the smoothie market are incurring economic loss; firms have an incentive to exit the market. By exiting the market these firms can cut down their losses and can utilize their resources in the new line of business.

Comment

Step **7** of 8

A perfectly competitive market is in long-run equilibrium when marginal cost (*MC*) is equal to the minimum of average total cost (*ATC*). As above table (b) shows, when output is 8 smoothies per hour, marginal cost is equal to the average total cost and average total cost is at its minimum ($4.25 per smoothie). In perfect competition, marginal cost (*MC*)

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Step-by-step solution

Step 1 of 8

(a) New technology generally allows firms to produce at a lower cost. So, firms adopting new technology experience a downward shift in cost curve.

Similarly technological change in Blu-ray production will allow firms to produce Blu-ray at lower cost. Firms producing Blu-ray player will experience a downward shift in their cost curves. At lower cost, these firms would be willing to supply more. As market supply increases, market supply curve will shift rightwards. With demand remaining same, this increase in supply will lead to fall in price of Blu-ray players. Firms that take first mover advantage in the production of Blu-ray player with new technology will reap economic profit while those which still produce with older technology will incur economic loss at lower prices.

Comment

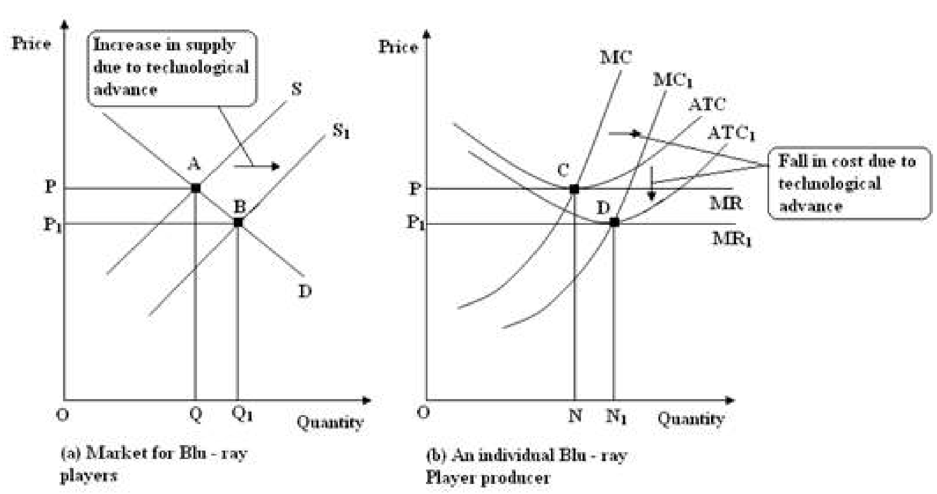
Step 2 of 8

These old technology firms will either exit the market or will adopt the new technology of producing Blu-ray players. With more new-technology firms entering the Blu-ray player market, price of Blu-ray players will fall and quantity produced will rise. Eventually, all firms that are using the new technology will remain in market and Blu-ray market will be in long-run equilibrium at a lower price and where all Blu-ray producers will be using the new technology and will be earning a zero economic profit.

Comment

Step 3 of 8

Following figure shows the Blu-ray market before and after technological change:



Comment

Step 4 of 8

Initially, in panel (a), market demand curve for Blu-ray players D intersects market supply curve of Blu-ray players S at point A and short run price *OP* is determined. Equilibrium quantity is *OQ*. This price, P, is equal to the long run average total cost of the firm as well. So, this price P is also the long-run price of Blu-ray players corresponding to supply conditions as presented by supply curve S.

At price OP, an individual Blu-ray players' producing firm is in equilibrium at point C and it is producing ON quantity of Blu-ray players {panel (b)}.

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Step 5 of 8

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New supply curve S_1 intersects the demand curve D at point B. New equilibrium price is OP_1. Equilibrium price has fallen from OP to OP_1. New equilibrium quantity is OQ_1. Equilibrium quantity has increased from OQ to OQ_1.

Comment

Step 6 of 8

New technology has allowed the firms to produce the Blu-ray players at lower cost. So, cost curves of firms have also shifted downwards from MC to MC_1 and ATC to ATC_1.

At new equilibrium price OP_1, individual Blu-ray player's producer is in equilibrium at point D and is producing ON_1 quantity of Blu-ray players. New price is also equal to the minimum total average total cost of the firm as shown by the intersection of MR_1 curve and ATC_1 curve. So, price OP_1, is also the long-run price corresponding to the supply conditions represented by new supply curve S_1.

Thus, technological change has resulted in lower price of Blu-ray Player in the long-run.

Comment

Step 7 of 8

(b) It has been stated that due to improvement in technology and mass production, cost of Blu-Ray player will fall to $90 per player in the long-run. Similar logic can be applied to the *HD VMD* as well. With continuous improvement of technology and mass production, cost of a *HD VMD* player will also fall because improvement in technology will lower the cost of production of *HD VMD* and mass production will increase the supply of *HD VMD*. Both of which in combination will lead to decline in prices of *HD VMD*.

Comment

Step 8 of 8

If the technological improvement and mass production of both the players run parallel to each other then, by virtue of being less expensive at the start, *HD VMD* still end up being less expensive in the long-run as well.

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Step-by-step solution

Step 1 of 5

In **Perfect Competition**, every firm maximizes its profit by producing the output where its marginal cost is equal to marginal revenue. In perfect competition, **Marginal Revenue is Equal to Price**. So, profit maximizing output can be attained at the level where the $\mathrm{P} = \mathrm{MC}$.

Comment

Step 2 of 5

Since *MC* curve of a firm above its average variable cost also acts as supply curve. So, through *MC* curve, we would be able to find the profit maximizing output at each given price. Thus, when firm is on its supply curve, it gets maximum value out of its resources because it is maximizing its profit. Summation of individual firms' supply curve provides the market supply curve and acts as marginal social cost curve.

Comment

Step 3 of 5

On the other hand, consumers also utilize their limited income in such a way that they can get maximum possible value out of it. Consumer's demand curve is the summation of all points at which consumer is allocating his budget in most efficient manner. So when consumer is on any point on his demand curve he is getting most out of his limited income. Summation of individual consumers' demand curve provides the market demand curve and acts as marginal social benefit curve.

Comment

Step 4 of 5

In **Perfect Competition**, even in short-run, equilibrium is attained at the intersection of market demand curve and market supply curve. The point of intersection lies on both demand curve and supply curve. So, market supply equals market demand at this point or we can say at point of intersection, marginal social cost equals the marginal social benefit and when $\mathrm{MSC} = \mathrm{MSB}$, resources are used efficiently. So, short-run equilibrium is efficient.

Comment

Step 5 of 5

Moreover, even if there is economic loss or economic profit will induce exit or entry into the market which eventually will bring the market to long-run equilibrium and market would be efficient.

So, until $\mathrm{MSB} = \mathrm{MSC}$, market is efficient even in short-run regardless of whether there is economic loss or economic profit.

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Step-by-step solution

Step 1 of 9

a.

At present time new technology and digitalization is in boom. Talking about the mobile sectors development is in boom. Higher percentage of consumer possesses android phones and phones that have different variants of operating systems for mobiles.

As on page 290, the market for smart phone apps is described. It is showing the opportunity in the app market since 2008. Market for app is less costly and the margin is very high. The features for the market of the app are as follows:

Comment

Step 2 of 9

1.

Mostly apps are made for the purpose of games and education. This is what market is delivering according to demand in the market. Maximum apps of these kinds are free to consumers. For the most apps, the common price is 99 cents. This means, there is a fixed price for the apps. It is one of the features of perfect competition.

2.

Availability of app developers in the market is easy. This means, there are large number of sellers and buyers in this market.

3.

The cost of making the app comes down as the demand is high and supply is not sufficient. This increases the profitability of the firms and entry of new firms starts. Increased demand and low cost of hiring the developers shows the higher margin in the market. This starts the entry of new firms in the market and so the competition is emerging.

Comment

Step 3 of 9

b.

"When the economic profit is eliminated from the market as a result of enough market supply, the market is said to be in long run equilibrium."

As per the article it can easily be understood that the demand for the apps are much higher. Even entry of new firms does not fulfill the demand in the market, because of the shortage in supply. The entry of new developers is still in continuation that indicates that there is economic profit in the market.

Therefore, the market for apps is not in long run equilibrium.

Comment

Step 4 of 9

c.

Advancement in technology lowers the developers' cost, because of the lower input firms can achieve higher output. As a result, the supply will increase that will make the market prices lower. Marginal revenue of developers will be equal to the market price.

Therefore, marginal revenue will be lowered as the prices will come down.

Another effect of technology advancement is the lower marginal cost of developers. The total cost, marginal cost and average cost curves will also shift downwards.

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Since, the production by the firms is at the level where marginal revenue is equal to the marginal cost, as a result of new technology advancement the marginal cost will decrease.

Therefore, in short run firms will be in win-win situation where the economic profit will be at maximum. In long run, firms would not be able to make enough economic profit, because the new technology will be used by all the firms in the market. Ultimately the competition will be emerged and the economic profit will come to zero (i.e. market will reach at equilibrium point).

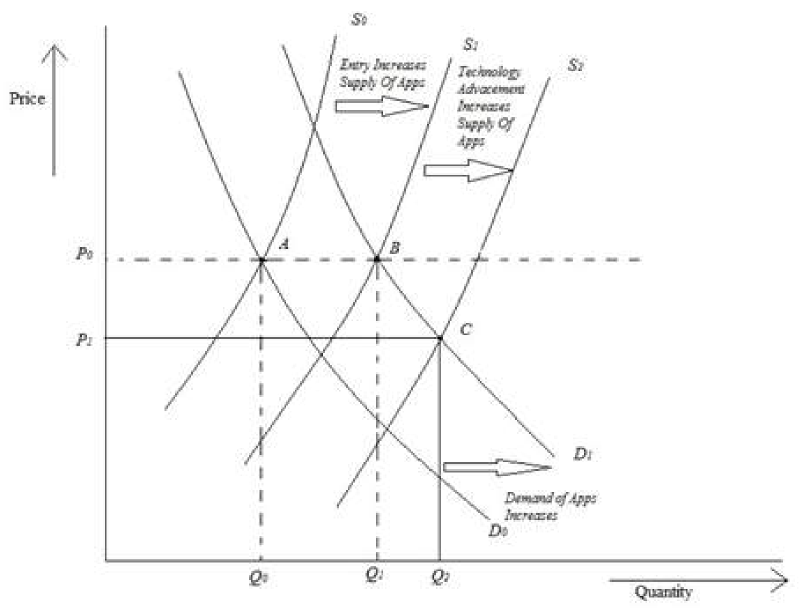
Comment

Step 6 of 9

d.

The Graph representing the change in supply as a result of technology advancement is as follows:

Figure 1.1:



Point ‘A’ represents the supply and demand in the market for app and the firms are producing the quantity Q_0. Increase in the margin for the app market encourages new

Comment

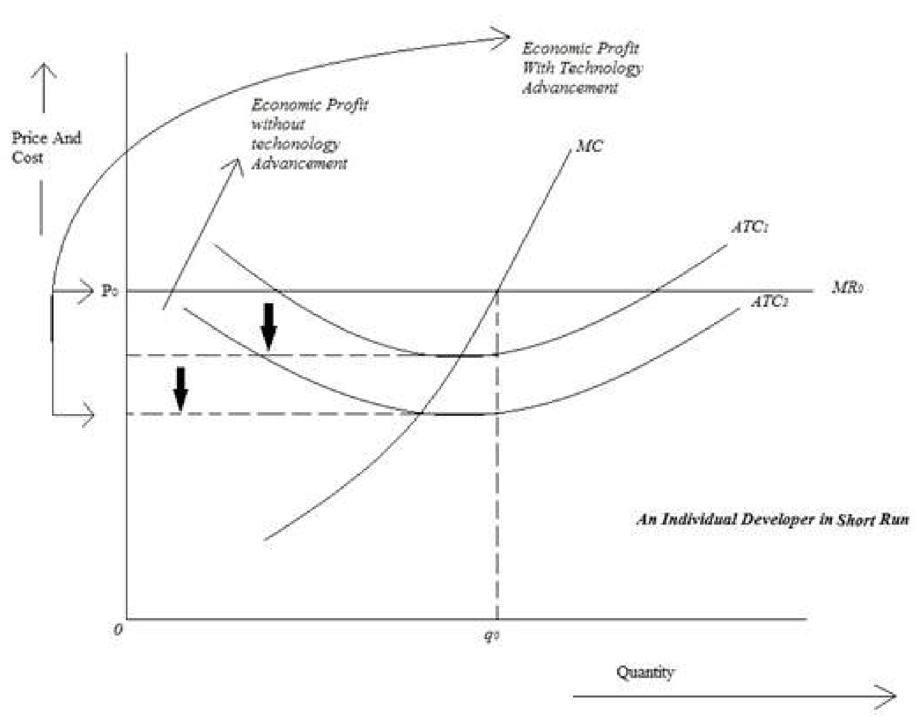
Step 7 of 9

firms to enter and thus the supply increases to S_1 full fill the demand D_1 and point ‘B’ is obtained.

Advancement in technology increases the supply more than that of earlier on the same level of demand and thus point ‘C’ is obtained with quantity of Q_2. Thus the new price level P_1 is obtained.

The graph representing the individual app developer in the short run is as follows:

Figure 1.2:



Comment

Step 8 of 9

When there is no technology advancement, the quantity is q_0 at the price level of P_0 with marginal revenue MR_0. Marginal revenue cut the marginal cost at the point where level of

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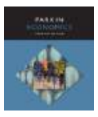
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Step-by-step solution

Step 1 of 9

a.

As the demand will increase, the firms will tend to maximize the profit by efficient allocation of the resources. The increased demand for the Smartphone indicates the consumer choice. Consumers value most to the smart phones. Therefore, producers will allocate all their resources to produce more of the smart phones. In short run, the profit will be maximized by the firm and the market will tend to achieve equilibrium point.

Increased global demand for smart phones will impact the market in a way that the demand will be equal to marginal social benefit for consumers. Supply from the producers is short in comparison to demand, so the supply will increase slightly compared to demand. Supply will be equal to marginal social cost for producers.

Comment

Step 2 of 9

Effect on Market:

After allocation of the resources the marginal social cost will increase slightly compared to marginal social benefit. This will increase the prices and producer surplus will be slightly higher than earlier. Efficient allocation is yet to be achieved as the case is in short run.

Therefore, as a result of increase in global demand for smart phones, the market will come at new point where the prices are higher and the producer surplus will also increase.

Comment

Step 3 of 9

Effect on individual Smartphone producer:

As a result of increased demand for the Smartphone, firms will experience the economic profit by using maximum of their resources in production. Increased demand will increase production for the firm and firm will experience economies of scale. This will lower the average total cost, marginal cost of production.

Other effect of increased demand will be the increase in the prices of the smart phones. There are few competitors and the supply is limited, shortage of supply increases the price of the product and thus the profit is maximized.

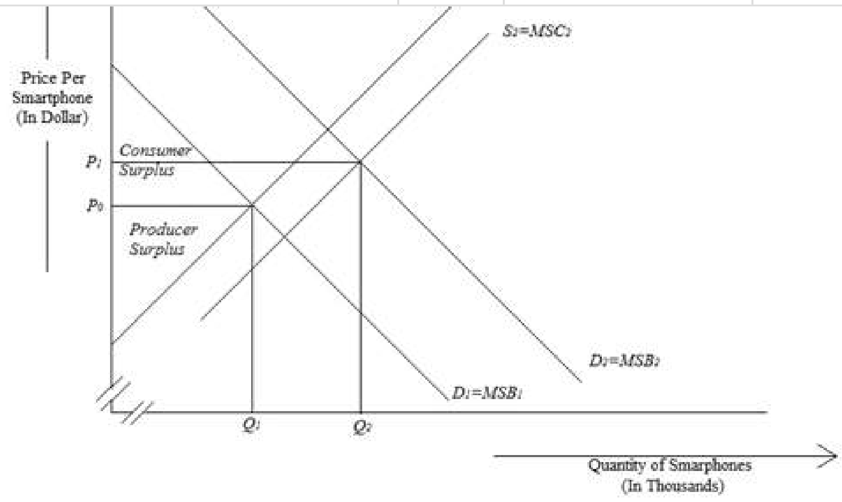
Comment

Step 4 of 9

b.

Graph representing the effect on market for Smartphones as global demand increases:

Figure 1.1:



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Step 5 of 9

The above graph represents the effect on the market for Smartphones when the global demand for phones increases. At the initial level when demand was not increased the market price for Smartphones was P_1 and Q_1 quantity was produced. The price and the quantity of production are achieved by the intersection of demand and supply curves.

Increased global demand pushed the demand curve rightwards and curves D_2 is obtained. This increased demand gets the attraction of the firms to reallocate the resources for the production. As a result of this action the supply increased and the new supply curve S_2 is obtained.

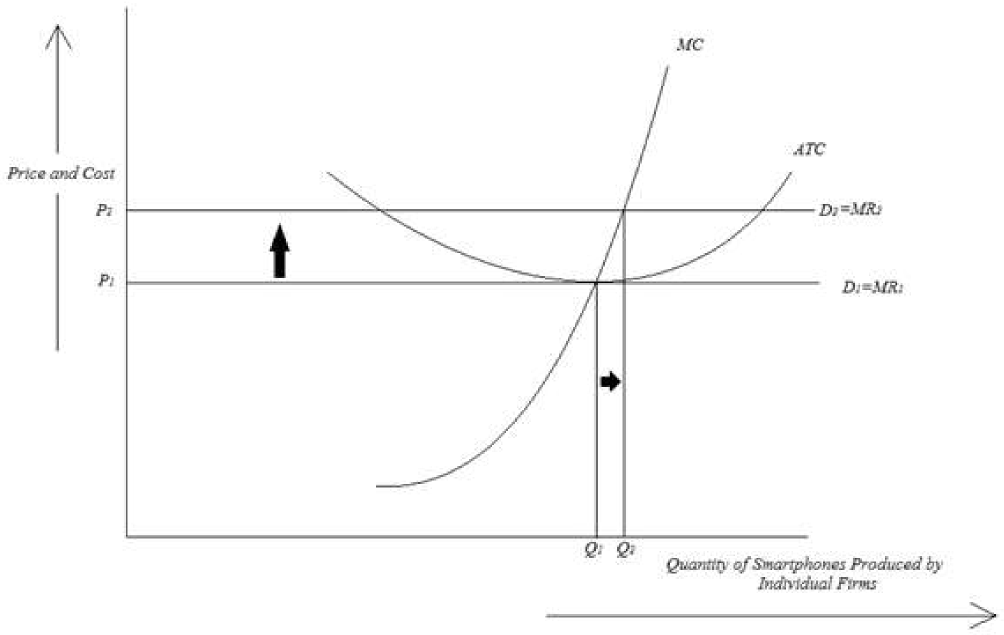
The new price level P_2 is achieved with increased level of production of quantity Q_2.

Comment

Step 6 of 9

Graph representing the effect on individual Smartphone Producer as global demand increases:

Figure 1.2:



Comment

Step 7 of 9

At the initial level the economic profit was almost zero. Increased demand enables individual firms to increase the factor of production.

In the above graph curve ATC is representing average total cost; curve MR is representing Marginal Revenue; curve MC is representing the marginal cost. Price P_1 is initial price with quantity of Q_1. Price is obtained with the intersection of MR curve and MC curve.

As the demand increases, the shortage of supply increases the price level and the firms decided to reallocate their resources to fulfill the demand. But firms are not able to achieve the production with minimum level of average total cost. The result comes by increased production of Q_2 quantity.

By these incident the new price level is achieved and the arrow between D_1 and D_2 is the economic profit that individual firm is enjoying in the short run.

c.

It can easily be understood that the price of the Smartphones is increased. This increased price level maximizes the economic profit. This economic profit is attracting the new entrants in the market. This is what is happening as per the news. Countries like 'Chi' and 'Ind' is emerging as the new market for the production of Smartphones at feasible prices. This is increasing the competition in the market.

Comment

Step 9 of 9

Higher economic profit will now be divided among the new firms those are entering. Existing firms will be compelled to cut down their prices in order to maintain the demand. This price cut as a result cuts the economic profit.

The entry of new firms in the future will further cut the economic profit and as a result of this a point will come where the economic profit will be zero.

Therefore in long run firms will achieve zero economic profit and will be able to produce the phones approximately at the lower average cost.

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